

Block cipher modes of operations

Module 2

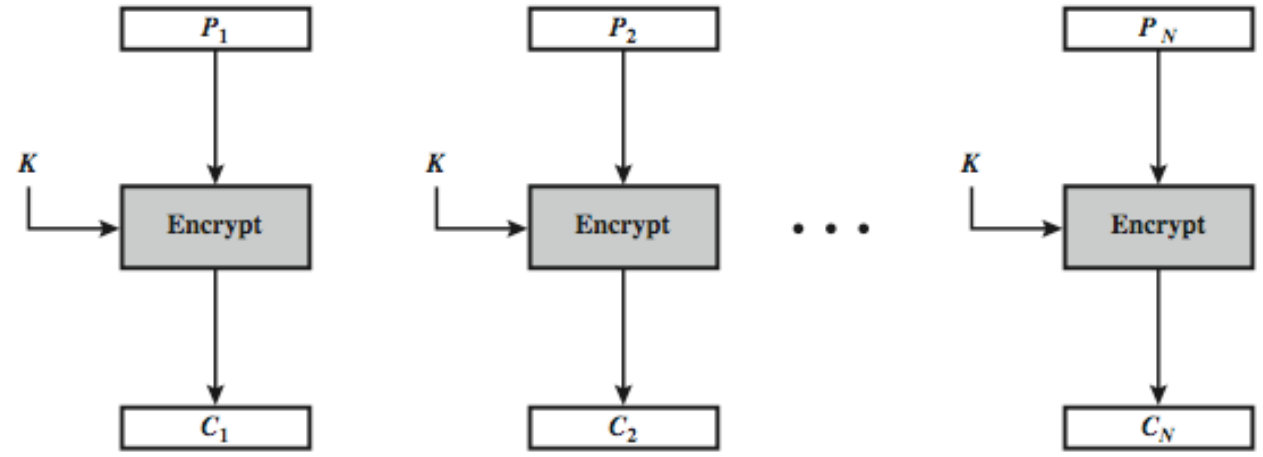
Modes of Operation

- block ciphers encrypt fixed size blocks
- eg. DES encrypts 64-bit blocks with 56-bit key
- Need some way to en/decrypt arbitrary amounts of data in practice
- NIST SP 800-38A defines 5 modes
- Have block and stream modes
- To cover a wide variety of applications
- Can be used with any block cipher

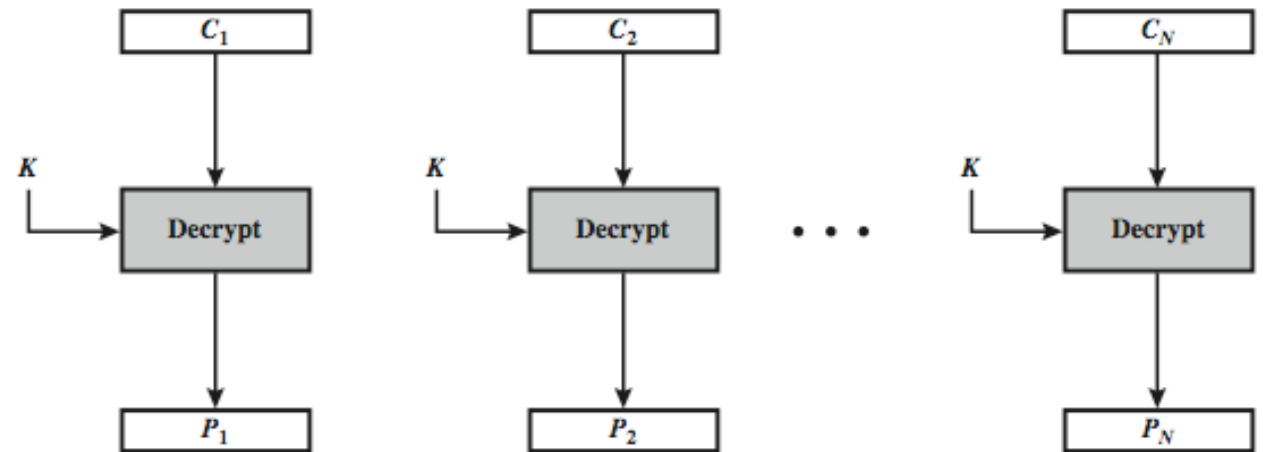
Electronic Codebook Book (ECB)

- Message is broken into independent blocks which are encrypted
- Each block is a value which is substituted, like a codebook, hence name
- Each block is encoded independently of the other blocks
$$C_i = E_K(P_i)$$
- uses: secure transmission of single values.

Electronic Codebook Book (ECB)



(a) Encryption



(b) Decryption

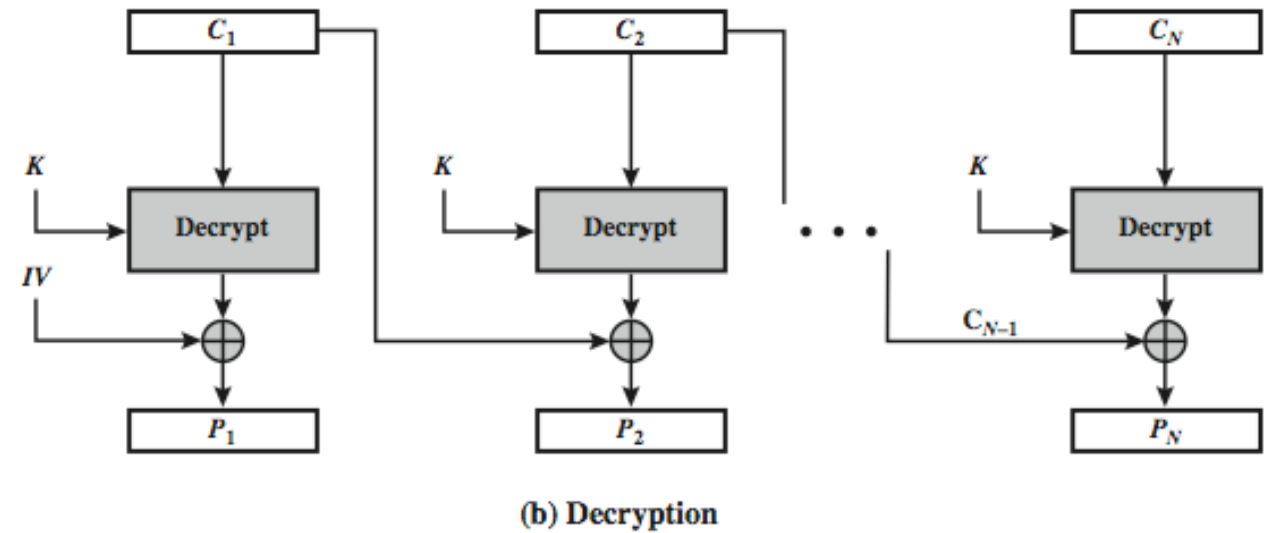
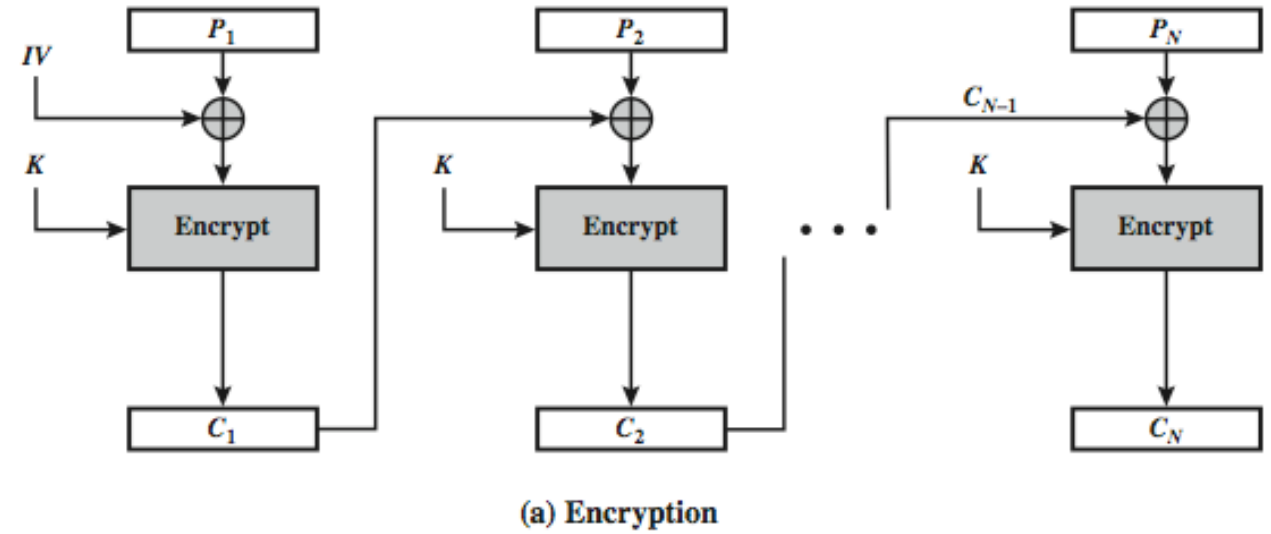
Advantages and limitations (ECB)

- Message repetitions may show in ciphertext.
- Weakness due to encrypted blocks being Independent.
- Main use of sending a few blocks of data.

Cipher Block Chaining (CBC)

- Message is broken into blocks
- Linked together in encryption operation
- Each previous cipher blocks is chained with current plaintext block, hence name.
- Use Initial Vector (IV) to start process
$$C_i = E_K(P_i \text{ XOR } C_{i-1})$$
$$C_{-1} = IV$$
- Uses: bulk data encryption, authentication

Cipher Block Chaining (CBC)



Message Padding

- At end of message must handle a possible last short block
 - which is not as large as blocksize of cipher
 - pad either with known non-data value (eg nulls)
 - or pad last block along with count of pad size
 - eg. [b1 b2 b3 0 0 0 0 5]
 - means have 3 data bytes, then 5 bytes pad+count
- this may require an extra entire block over those in message

Advantages and limitations of CBC

- A ciphertext block depends on **all** blocks before it
- Any change to a block affects all following ciphertext blocks
- Need **Initialization Vector (IV)**
 - which must be known to sender & receiver
 - if sent in clear, attacker can change bits of first block, and change IV to compensate
 - hence IV must either be a fixed value (as in EFTPOS)
 - or must be sent encrypted in ECB mode before rest of message

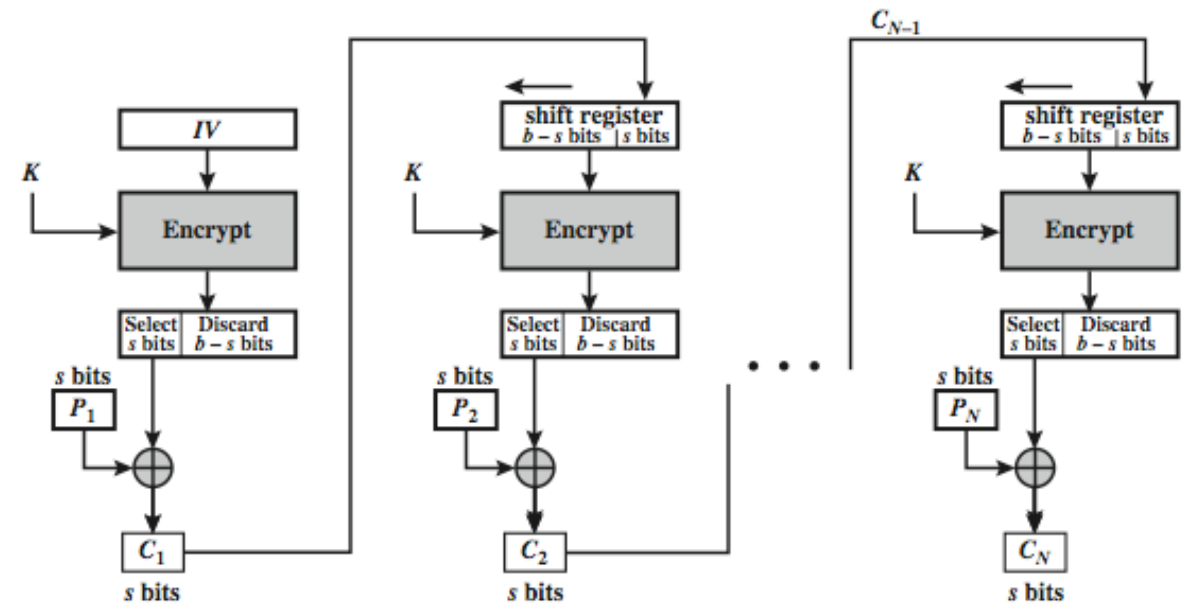
Stream Modes of Operation

- block modes encrypt entire block
- May need to operate on smaller units
 - real time data
- Convert block cipher into stream cipher
 - cipher feedback (CFB) mode
 - output feedback (OFB) mode
 - counter (CTR) mode

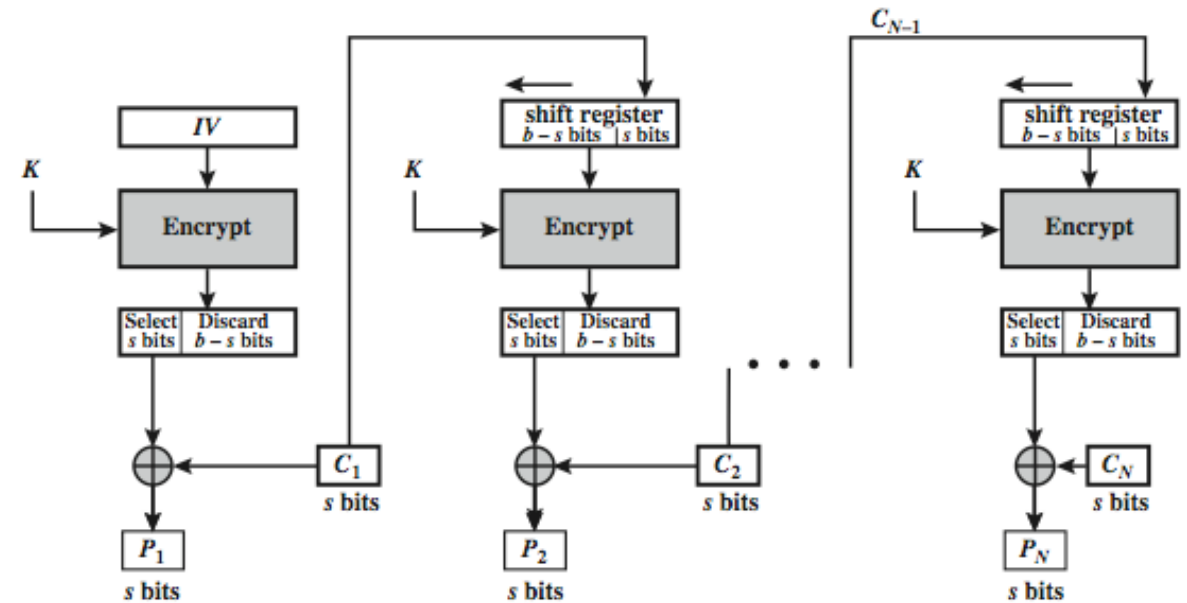
Cipher FeedBack (CFB)

- Message is treated as a stream of bits
- Added to the output of the block cipher
- Result is feed back for next stage (hence name)
- Standard allows any number of bit (1,8, 64 or 128 etc) to be feed back
 - denoted CFB-1, CFB-8, CFB-64, CFB-128 etc
- Most efficient to use all bits in block (64 or 128)
$$C_i = P_i \text{ XOR } E_K(C_{i-1})$$
$$C_{-1} = IV$$
- Uses: stream data encryption, authentication

s-bit Cipher FeedBack (CFB-s)



(a) Encryption



(b) Decryption

Advantages and Limitations of CFB

- Appropriate when data arrives in bits/bytes
- Most common stream mode
- Limitation is need to stall while do block encryption after every n-bits
- Note that the block cipher is used in encryption mode at both ends
- Errors propagate for several blocks after the error

Output FeedBack (OFB)

- Message is treated as a stream of bits
- Output of cipher is added to message
- Output is then feed back (hence name)
- Feedback is independent of message
- Can be computed in advance

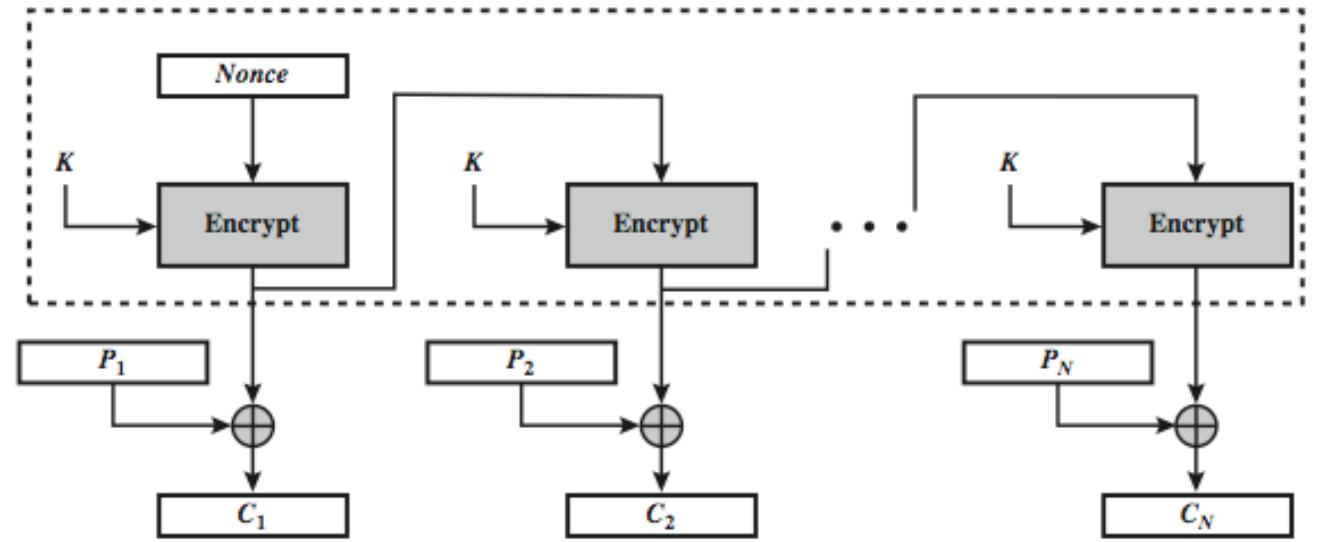
$$O_i = E_K(O_{i-1})$$

$$C_i = P_i \text{ XOR } O_i$$

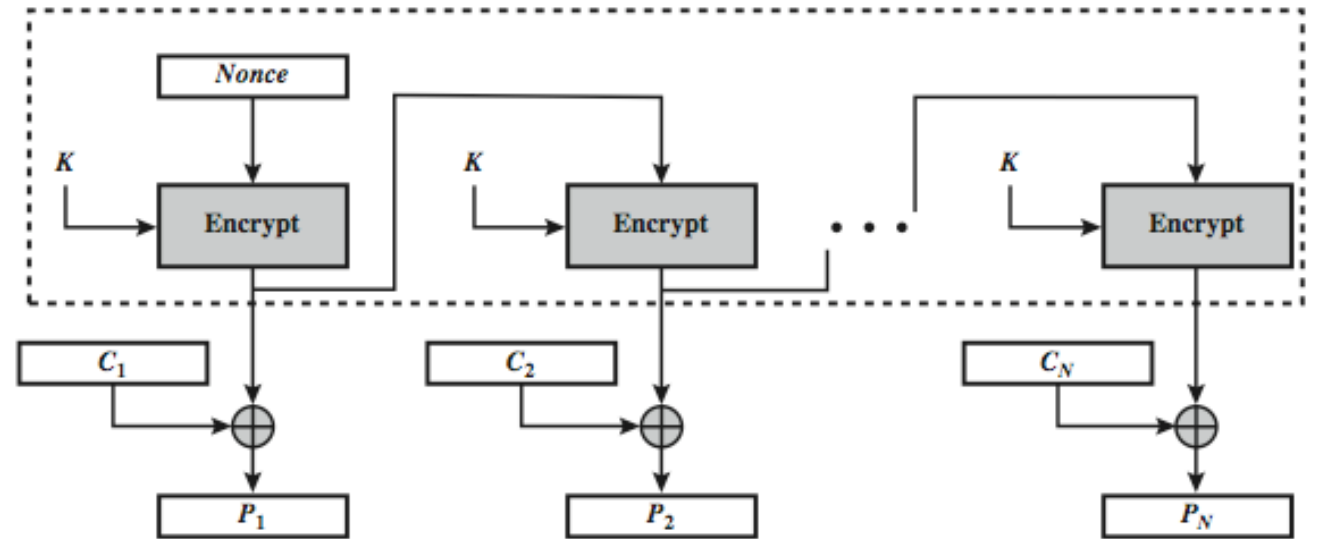
$$O_{-1} = IV$$

- Uses: stream encryption on noisy channels

Output FeedBack (OFB)



(a) Encryption



(b) Decryption

Advantages and Limitations of OFB

- needs an IV which is unique for each use
 - if ever reuse attacker can recover outputs
- bit errors do not propagate
- more vulnerable to message stream modification
- sender & receiver must remain in sync
- only use with full block feedback
 - subsequent research has shown that only **full block feedback** (ie CFB-64 or CFB-128) should ever be used

Counter (CTR)

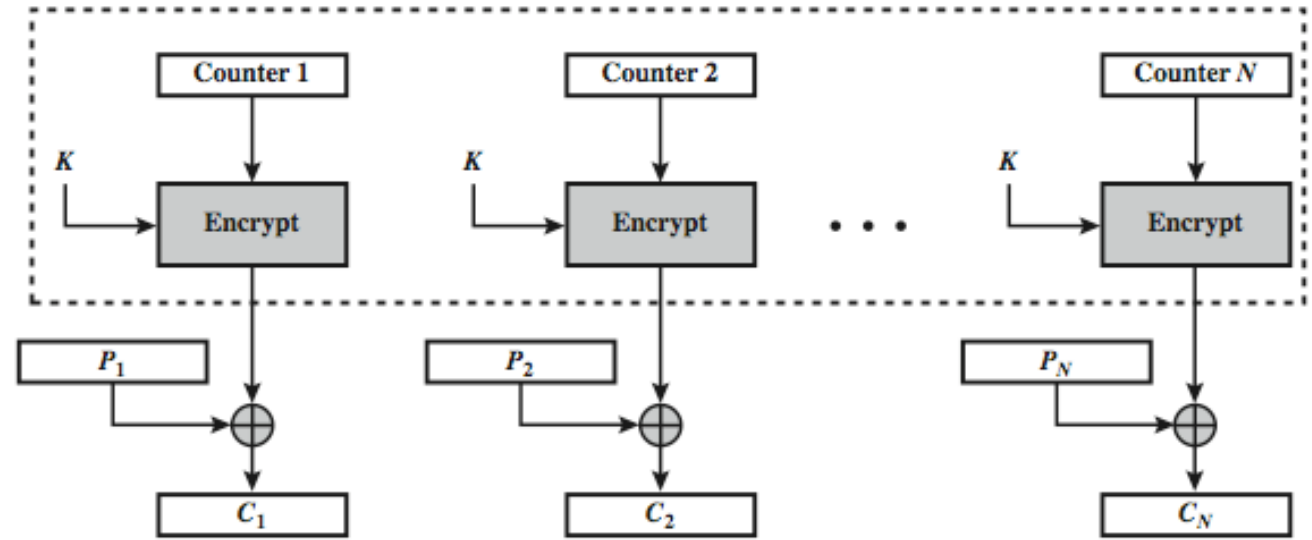
- a “new” mode, though proposed early on
- similar to OFB but encrypts counter value rather than any feedback value
- must have a different key & counter value for every plaintext block (never reused)

$$O_i = E_K(i)$$

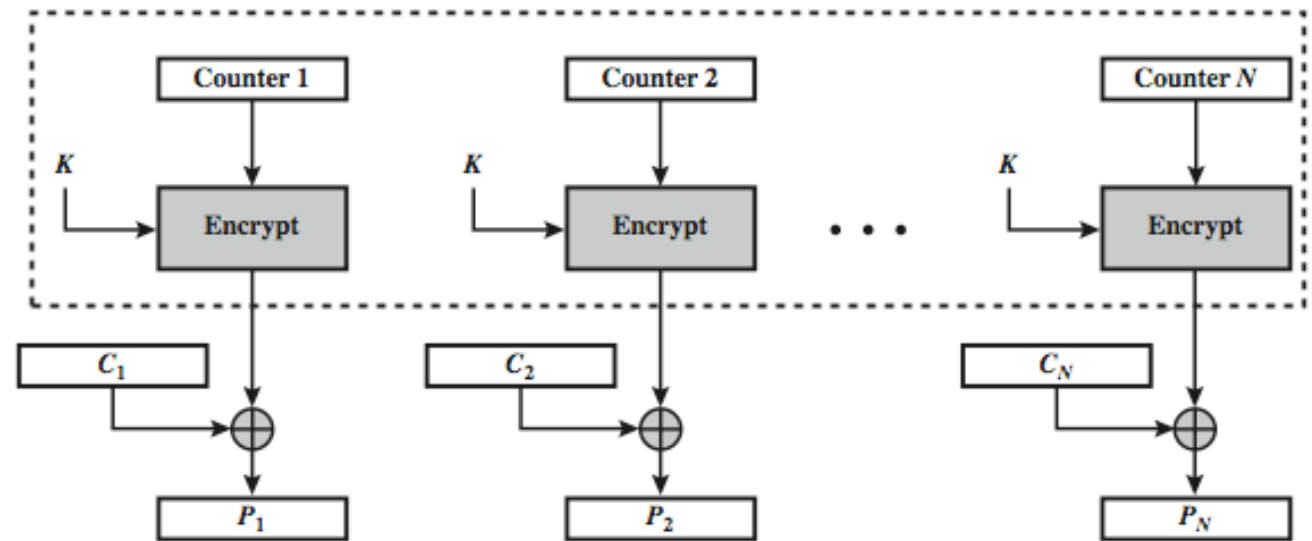
$$C_i = P_i \text{ XOR } O_i$$

- uses: high-speed network encryptions

Counter (CTR)



(a) Encryption

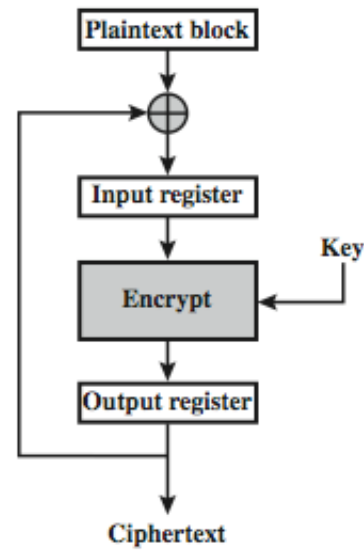


(b) Decryption

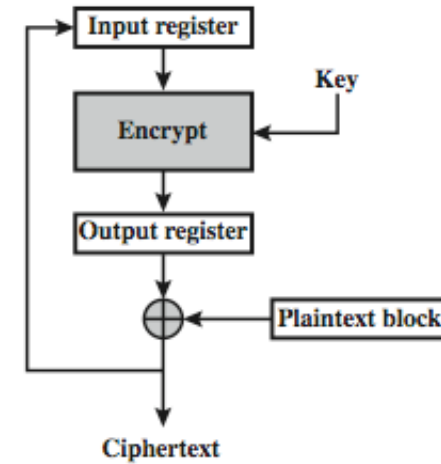
Advantages and Limitations of CTR

- Efficiency
 - can do parallel encryptions in h/w or s/w
 - can preprocess in advance of need
 - good for bursty high speed links
- Random access to encrypted data blocks
- Provable security (good as other modes)
- But must ensure never reuse key/counter values, otherwise could break (cf OFB)

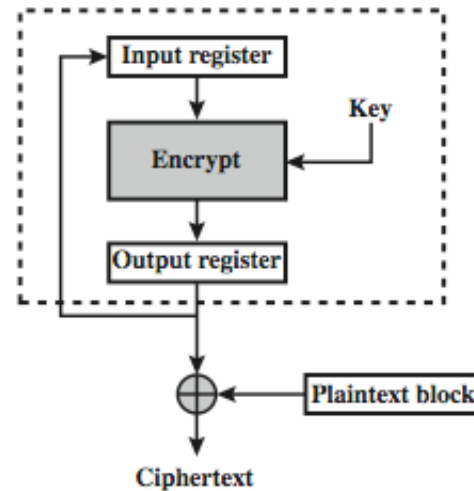
Feedback Character- istics



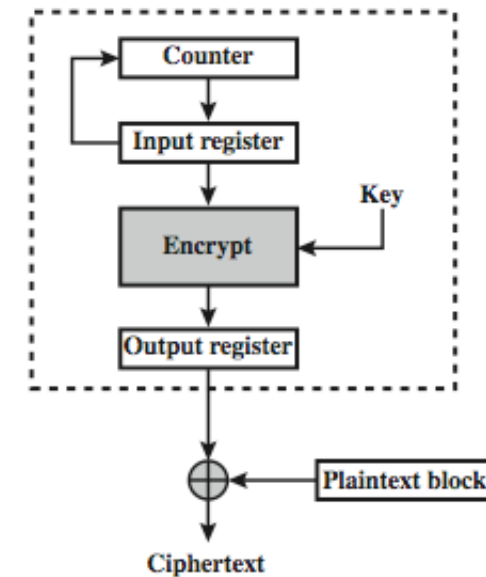
(a) Cipher block chaining (CBC) mode



(b) Cipher feedback (CFB) mode



(c) Output feedback (OFB) mode



(d) Counter (CTR) mode